Chapter 2: Atomic Structure & Bonding

Key Equations

Average Atomic Weight: $A_{avg} = \sum f_i A_i$ (f_i: fraction of occurrence)

Force of Attraction: $F_A = \frac{|Z_1 Z_2| e^2}{4\pi\epsilon_0 r^2}$

(Z_1, Z_2: ion valences, r: interatomic distance, $e = 1.602 \times 10^{-19}$ C, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m)

% Ionic Character: $\%IC = \left(1 - e^{-0.25(X_A - X_B)^2}\right) \times 100$

(X_A, X_B: electronegativities)

Concepts

Bohr Model: Electrons orbit nucleus in discrete energy levels ($E = hf$).

Bonding: *Ionic*—electron transfer. *Covalent*—electron sharing. *Metallic*—sea of delocalized electrons.

Chapter 3: Crystal Structures

Key Equations & Geometry

SC: $a = 2R$, 1 atom/cell. **BCC:** $a = \frac{4R}{\sqrt{3}}$, 2 atoms/cell.

FCC: $a = 2R\sqrt{2}$, 4 atoms/cell.

APF: $\frac{\text{Vol. of atoms}}{\text{Vol. of unit cell}} = \frac{n(\frac{4}{3}\pi R^3)}{a^3}$ (SC=0.52, BCC=0.68, FCC=0.74)

Theoretical Density: $\rho = \frac{nA}{V_c N_A}$ ($N_A = 6.022 \times 10^{23}$ atoms/mol)

Miller Indices (Planes): (1) If plane passes through origin, establish new origin. (2) Find intercepts on x, y, z. (3) Take reciprocals. (4) Clear fractions. (5) Enclose in (hkl); negatives get a bar.

Concepts

Isotropy/Anisotropy: Isotropic—properties independent of direction; anisotropic—vary with direction.

Chapter 4: Imperfections in Solids

Key Equations

Total Atomic Sites: $N = \frac{N_A \rho}{A}$

Equilibrium Vacancies: $N_v = N \exp\left(-\frac{Q_v}{kT}\right)$ ($k = 8.62 \times 10^{-5}$ eV/atom·K)

Mass to Weight %: $C_1 = \frac{m_1}{m_1 + m_2} \times 100$

Weight % to Atom %: $C_1' = \frac{C_1/A_1}{C_1/A_1 + C_2/A_2} \times 100$

Atom % to Weight %: $C_1 = \frac{C_1' A_1}{C_1' A_1 + C_2' A_2} \times 100$

Concepts

Vacancy: Atom missing from lattice site. **Self-Interstitial:** Extra atom in void space.

Chapter 5: Diffusion

Key Equations

Fick's First Law (Steady-State): $J = -D\frac{\Delta C}{\Delta x}$

Mass Diffused: $M = JAt = -DAt\frac{\Delta C}{\Delta x}$

Fick's Second Law (Non-Steady-State):

$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$

Diffusion Coefficient: $D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$ ($R = 8.31$ J/mol·K)

Activation Energy (Two Temps): $Q_d = -R\frac{\ln D_1 - \ln D_2}{1/T_1 - 1/T_2}$

erf Table & Interpolation: $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$

z	erf	z	erf	z	erf	z	erf
0.00	0.000	0.30	0.329	0.60	0.604	1.00	0.843
0.10	0.112	0.40	0.428	0.70	0.678	1.20	0.910
0.20	0.223	0.50	0.521	0.80	0.742	1.50	0.966
0.25	0.276	0.55	0.563	0.90	0.797	2.00	0.995

Chapter 6: Mechanical Properties

Key Equations

Eng. Stress: $\sigma = F/A_0$ **Eng. Strain:** $\epsilon = \Delta l / l_0$

Shear: $\tau = F/A_0$, $\gamma = \tan\theta$

Hooke's Law: $\sigma = E\epsilon$ (elastic only) $\tau = G\gamma$

Poisson's Ratio: $\nu = -\epsilon_x/\epsilon_z = -\epsilon_y/\epsilon_z$

True Stress/Strain: $\sigma_T = \sigma(1 + \epsilon)$; $\epsilon_T = \ln(1 + \epsilon)$

Strain Hardening: $\sigma_T = K\epsilon_T^n$

Resilience: $U_r \approx \frac{1}{2}\sigma_y \epsilon_y = \frac{\sigma_y^2}{2E}$

Elastic Elongation: $\Delta l = \frac{l_0 F}{A_0 E}$

Concepts

Yield Strength: 0.002 strain offset. **Tensile Strength:** Max on engineering curve. **% Elongation:** $\frac{l_f - l_0}{l_0} \times 100$.

Chapter 7: Dislocations & Strengthening

Key Equations

Resolved Shear Stress: $\tau_R = \sigma \cos\phi \cos\lambda$

Yield Strength: $\sigma_y = \frac{\tau_{CRSS}}{(\cos\phi\cos\lambda)_{\max}}$

Hall-Petch: $\sigma_y = \sigma_0 + k_y d^{-1/2}$

% Cold Work: $\%CW = \left(\frac{A_0 - A_d}{A_0}\right) \times 100$

Concepts

Slip System: Slip plane + slip direction. **Strain Hardening:** Ductile metal gets harder/stronger as plastically deformed (more dislocations).

Chapter 8: Failure

Key Equations

Stress Concentration: $\sigma_m = 2\sigma_0\sqrt{a/\rho_t} = K_t\sigma_0$

Critical Stress (Brittle): $\sigma_c = \sqrt{\frac{2E\gamma_s}{\pi a}}$

Fracture Toughness: $K_c = Y\sigma_c\sqrt{\pi a}$

Cyclic: $\Delta\sigma = \sigma_{\max} - \sigma_{\min}$; $\sigma_m = (\sigma_{\max} + \sigma_{\min})/2$; $R = \sigma_{\min}/\sigma_{\max}$

Creep Rate: $\dot{\epsilon}_s = K_2\sigma^n \exp(-Q_c/RT)$

Larson-Miller: $m = T(C + \log t_r)$

Concepts

Fatigue: Structural damage under cyclic stress. **S-N Curve:** Fatigue lifetime = cycles to failure at given σ; fatigue strength = σ for given cycles.

Chapter 9: Phase Diagrams

Key Equations — Lever Rule

For a two-phase region with tie line from C_α to C_β at composition C_0:

$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$ $W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha}$

Eutectic Microconstituent: $W_e = \frac{C_0' - 18.3}{43.6}$ (Pb-Sn)

Fe-C Hypoeutectoid ($C_0' < 0.76$ wt%C):

$W_p = \frac{C_0' - 0.022}{0.74}$ $W_{\alpha'} = \frac{0.76 - C_0'}{0.74}$

Fe-C Hypereutectoid ($C_0' > 0.76$ wt%C):

$W_p = \frac{6.70 - C_1'}{5.94}$ $W_{Fe_3C'} = \frac{C_1' - 0.76}{5.94}$

Fe-C Key Points

Eutectoid: $\gamma(0.76\%C) \to \alpha(0.022\%C) + Fe_3C(6.7\%C)$ at 727°C

Eutectic: $L \to \gamma + Fe_3C$ at 1147°C, 4.30 wt%C

Phases: α-ferrite (BCC, soft); austenite γ (FCC); cementite Fe_3C (hard)

Hypoeutectoid (<0.76%C): proeutectoid α + pearlite

Hypereutectoid (>0.76%C): proeutectoid Fe_3C + pearlite

Reactions (Definitions)

Eutectic: $L \to \alpha + \beta$ (liquid → two solids)

Eutectoid: $\gamma \to \alpha + Fe_3C$ (solid → two solids)

Peritectic: $L + \alpha \to \beta$ (liquid + solid → solid)

Example: Fe-C Lever Rule (HW7)

0.45 wt%C alloy (hypoeutectoid) at room temp:

$W_p = \frac{0.45 - 0.022}{0.74} = 0.578$ $W_{\alpha'} = \frac{0.76 - 0.45}{0.74} = 0.419$

Chapter 10: Phase Transformations

Key Equations

Critical Radius for Nucleation:
$r^* = -\frac{2\gamma T_m}{\Delta H_f} \cdot \frac{1}{T_m - T} = \frac{-2\gamma}{\Delta G_v}$
(γ: surface free energy, ΔH_f: latent heat of fusion, T_m: melting temp K, T: transformation temp K)
Activation Free Energy:
$\Delta G^* = \frac{16\pi\gamma^3 T_m^2}{3\Delta H_f^2} \cdot \frac{1}{(T_m - T)^2}$
Avrami Equation: $y = 1 - \exp(-kt^n)$
(y: fraction transformed, k, n: material constants)
Transformation Rate: rate $= 1/t_{0.5}$

Example: Critical Radius (HW9, 10.3)

Cu: $T_m = 1085°$C, nucleates at 849°C, $\Delta H_f = -1.77 \times 10^9$ J/m^3, $\gamma = 0.200$ J/m^2:
$r^* = \frac{-2(0.200)(1085+273)}{-1.77\times10^9} \cdot \frac{1}{1085-849} = 1.30 \times 10^{-9}$ m

Example: Avrami Rate (HW9, 10.8)

$n = 2.5$, $y = 0.40$ at $t = 200$ min. Find k: $k = \frac{-\ln(1-y)}{t^n} = \frac{-\ln(0.60)}{200^{2.5}} = 9.0 \times 10^{-8}$.
Find $t_{0.5}$: $t_{0.5} = \left[\frac{-\ln(0.5)}{k}\right]^{1/n} = \left[\frac{0.693}{9.0\times10^{-8}}\right]^{1/2.5} = 311$ min; rate $= 1/311 = 3.2 \times 10^{-3}$ min^{-1}

Chapter 12: Ceramics

Key Equations

Theoretical Density: $\rho = \frac{n'(\Sigma A_C + \Sigma A_A)}{V_C N_A}$
(n': formula units/cell, A_C, A_A: cation/anion atomic weights, V_C: unit cell volume)
Frenkel Defect #: $N_{fr} = N \exp\left(-\frac{Q_{fr}}{2kT}\right)$
Schottky Defect #: $N_s = N \exp\left(-\frac{Q_s}{2kT}\right)$

Concepts

Frenkel Defect: A cation vacancy + cation interstitial pair (cation moves from site to void).
Schottky Defect: A paired set of cation vacancy AND anion vacancy (maintains charge neutrality).

Example: Schottky (HW9, 12.29)

NaCl at 801°C ($T = 1074$ K), $Q_s = 2.3$ eV:
$N_s/N = \exp\left(\frac{-2.3}{2(8.62\times10^{-5})(1074)}\right) = \exp(-12.43) = 3.97 \times 10^{-6}$

Chapter 14: Polymers

Key Equations

Number-Avg Mol. Weight: $M_n = \sum x_i M_i$ (x_i: number fraction)
Weight-Avg Mol. Weight: $M_w = \sum w_i M_i$ (w_i: weight fraction)
Degree of Polymerization: $DP = M_n/m$ (m: repeat unit mol. wt)
Repeat Unit MW (copolymer): $m = \sum f_j m_j$ (f_j: mole fraction)

Example (HW10)

Polypropylene $M_n = 10^6$ g/mol, repeat unit C_3H_6:
$m = 3(12.01) + 6(1.01) = 42.09$ g/mol; $DP = 10^6/42.09 = 23,760$

Chapter 18: Electrical Properties

Key Equations

Ohm's Law: $V = IR$ **Resistivity:** $\rho = RA/l$ **Conductivity:** $\sigma = 1/\rho$
Current Density: $J = \sigma\mathcal{E}$ **E-field:** $\mathcal{E} = V/l$
Metal Conductivity: $\sigma = n|e|\mu_e$
Matthiessen's Rule: $\rho_{total} = \rho_t + \rho_i + \rho_d$
Intrinsic Semiconductor: $n = p = n_i$; $\sigma = n_i|e|(\mu_e + \mu_h)$
n-type extrinsic: $\sigma \cong n|e|\mu_e$ **p-type:** $\sigma \cong p|e|\mu_h$
Capacitance (vacuum): $C = \epsilon_0 A/l$ $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
Capacitance (dielectric): $C = \epsilon A/l = \epsilon_r\epsilon_0 A/l$
Charge: $C = Q/V$

Constants

$|e| = 1.6 \times 10^{-19}$ C; Si: $\mu_e = 0.14$ m^2/V·s, $\mu_h = 0.048$ m^2/V·s, $n_i = 1.1 \times 10^{16}$ m^{-3}
Ge: $\mu_e = 0.38$, $\mu_h = 0.18$ m^2/V·s, $n_i = 2.4 \times 10^{19}$ m^{-3}

Concepts

Intrinsic Semiconductor: Conductivity from pure material; $n = p$ (e.g., pure Si, Ge).
Extrinsic Semiconductor: Conductivity controlled by dopants.
n-type = donor impurity (Group V, e.g., P in Si), excess electrons.
p-type = acceptor impurity (Group III, e.g., B in Si), excess holes.

Example: Intrinsic Carrier Concentration (HW10)

GaAs: $\sigma = 3 \times 10^{-7}$ $(\Omega\cdot\text{m})^{-1}$, $\mu_e = 0.80$, $\mu_h = 0.04$ m^2/V·s:
$n_i = \frac{\sigma}{|e|(\mu_e+\mu_h)} = \frac{3\times10^{-7}}{(1.6\times10^{-19})(0.84)} = 2.2 \times 10^{12}$ m^{-3}

Example: Extrinsic Conductivity (HW10)

Ge doped with 5×10^{22} m^{-3} Sb (n-type), $\mu_e = 0.1$, $\mu_h = 0.05$ m^2/V·s:
$\sigma \cong n|e|\mu_e = (5 \times 10^{22})(1.6 \times 10^{-19})(0.1) = 800$ $(\Omega\cdot\text{m})^{-1}$

Example: Capacitance

Parallel plate, $A = 2 \times 10^{-4}$ m^2, $l = 1 \times 10^{-3}$ m, $\epsilon_r = 6.0$:
$C = (6.0)(8.85 \times 10^{-12})(2 \times 10^{-4})/(1 \times 10^{-3}) = 1.06 \times 10^{-12}$ F

Chapter 19: Thermal Properties

Key Equations

Heat Capacity: $C = dQ/dT$ (J/mol·K)
Specific Heat: $c = dQ/(m\,dT)$ (J/kg·K — per unit mass)
Linear Thermal Expansion: $\Delta l/l_0 = \alpha_l \Delta T$
Volume Expansion: $\Delta V/V_0 = \alpha_v \Delta T \approx 3\alpha_l \Delta T$
Heat Flux (Fourier's Law): $q = -k\,dT/dx$
Thermal Stress: $\sigma = E\alpha_l(T_0 - T_f)$

Concepts

Heat Capacity (C): Energy required to raise 1 mol of material by 1 K.
Specific Heat (c): Heat capacity per unit mass; same concept but normalized per kg. Measured at constant pressure (C_p) or volume (C_v); $C_p > C_v$.
Thermal Conductivity: Metals highest (free electrons), ceramics moderate (phonons), polymers lowest.
Thermal Shock: Fracture from rapid temp changes; $TSR \propto \sigma_f k/(E\alpha_l)$.

HW Problem Reference

Phase Diagram — Compositions at a Temperature (HW7)

Given a binary phase diagram at equilibrium, locate the tie line at the temperature. Read compositions of each phase from the phase boundaries, then use the lever rule to get mass fractions.

Resistivity Calculation (HW10)

$\rho = RA/l$. Example: wire of length l, cross-section $A = \pi r^2$, resistance R. Find ρ or $\sigma = 1/\rho$.
Example: $A = \pi(0.001)^2 = 3.14 \times 10^{-6}$ m^2, $R = 0.03$ V/A$= 0.003$ Ω per 300 mm: $\rho = RA/l = (0.003)(3.14 \times 10^{-6})/(0.3) = 3.14 \times 10^{-8}$ Ω·m.

Ceramic Density (HW9, 12.16)

Rock salt (NaCl): $n' = 4$ formula units/cell, $A_{Na} = 22.99$, $A_{Cl} = 35.45$, $a = 0.562$ nm:
$\rho = \frac{4(22.99+35.45)}{(0.562\times10^{-9})^3(6.022\times10^{23})} = 2.16 \times 10^3$ kg/m^3

Polymer MW (HW10)

$M_n = \sum x_i M_i$; $M_w = \sum w_i M_i$; $DP = M_n/m$. Use midpoint of each range as M_i.

Nucleation (HW9)

Solve for k from $y = 1 - \exp(-kt^n)$: rearrange to $k = \frac{-\ln(1-y)}{t^n}$. Then find $t_{0.5}$ from $\ln(0.5) = -kt_{0.5}^n$.

Common Constants

$R = 8.314$ J/mol·K, $k_B = 8.62 \times 10^{-5}$ eV/atom·K, $N_A = 6.022 \times 10^{23}$, $|e| = 1.6 \times 10^{-19}$ C, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m